Operating System Lab Observation

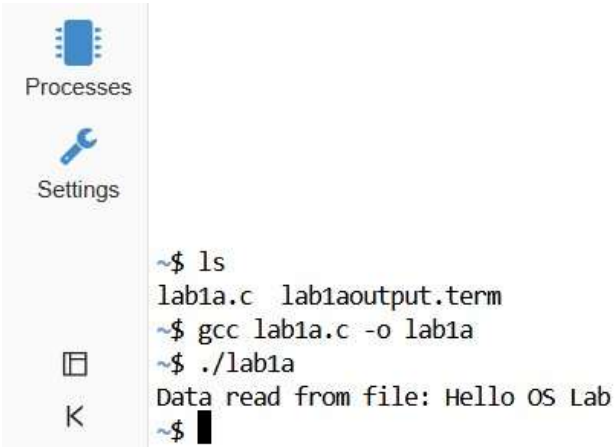
1a

Program:

```
#include <stdio.h>

#include <unistd.h>

#include <fcntl.h>


int main()

{    int

fd;

   char buffer[50];


   /* open() – create and open a file */     fd =

open("sample.txt", O_CREAT | O_RDWR, 0644);


   if (fd < 0)

   {

      printf("File opening failed\n");

return 1;

   }


   /* write() – write data into file */

write(fd, "Hello OS Lab", 12);


   /* move file pointer to beginning */

lseek(fd, 0, SEEK_SET);
```

```
   /* read() – read data from file */
read(fd, buffer, 12);

   /* display data */     printf("Data read from
file: %s\n", buffer);

   /* close() – close the file */
close(fd);

   return 0;
}
```

Output:

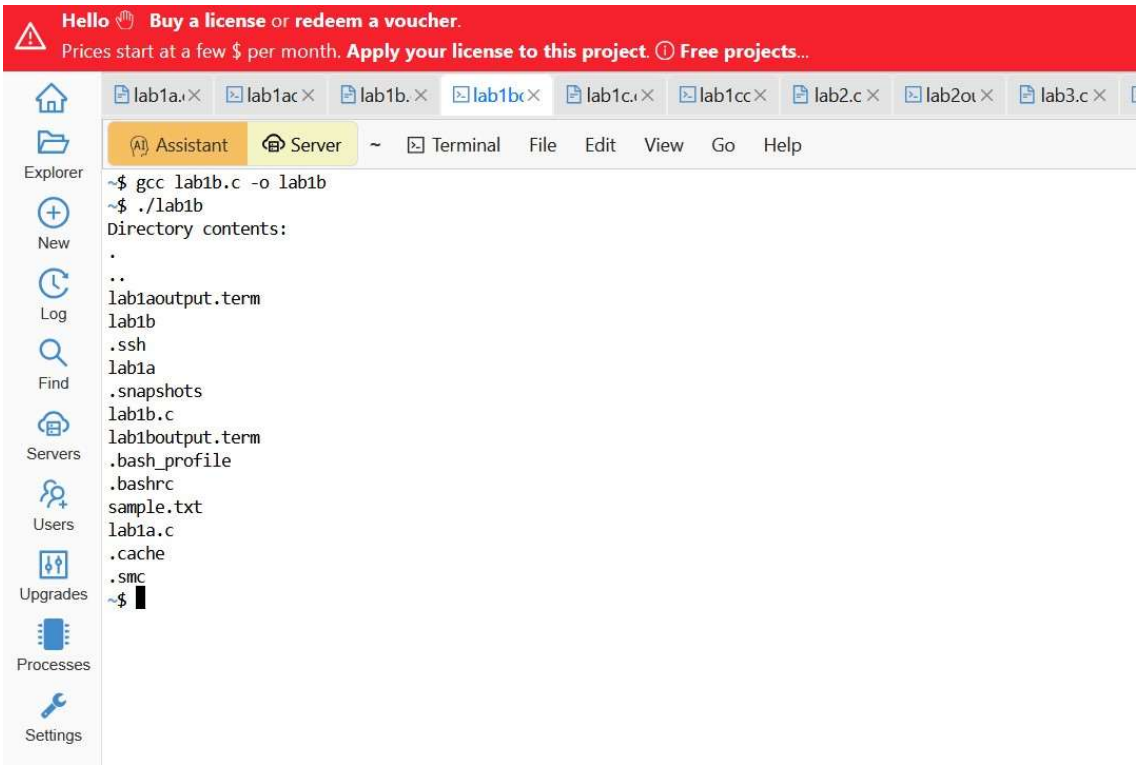


1b:

Program:

```
#include <stdio.h>
#include <dirent.h>

int main()
{
   DIR *d;     struct
dirent *entry;

   /* open current directory */
d = opendir(".");

   if (d == NULL)
   {
      printf("Directory cannot be opened\n");
      return 1;
   }
   /* read directory entries */
printf("Directory contents:\n");     while
((entry = readdir(d)) != NULL)
   {
      printf("%s\n", entry->d_name);
   }
   /* close directory */
closedir(d);     return
0;
}
```

Output:



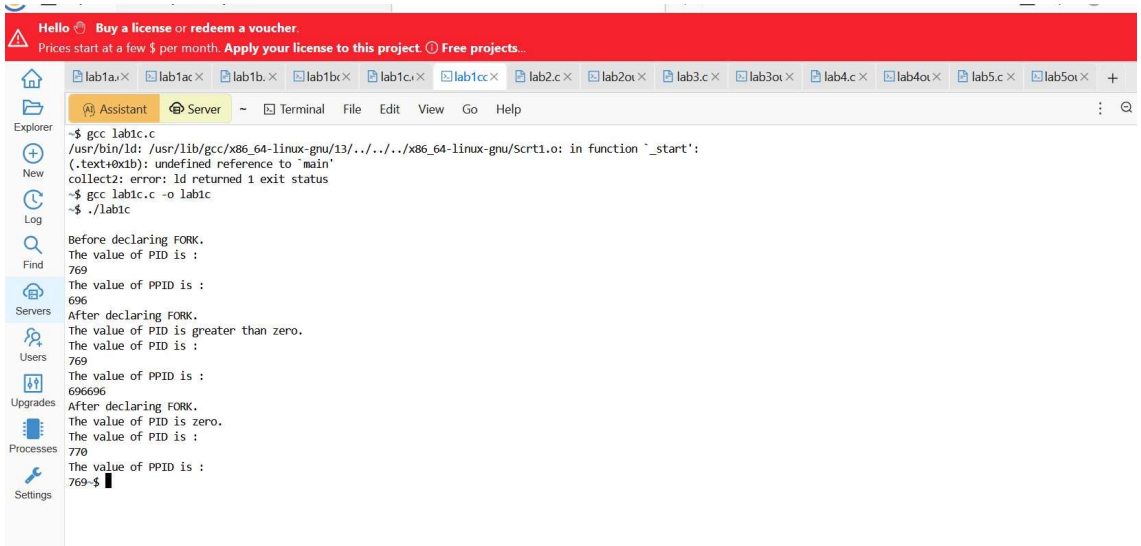
1c:

Program:

```
#include<stdio.h>

#include<unistd.h>

#include<sys/types.h> void

main()

{

printf("\nBefore declaring FORK.");

printf("\nThe value of PID is : ");

printf("\n%d",getpid()); printf("\nThe

value of PPID is : ");

printf("\n%d",getppid());


pid_t pid=fork();//Used to create child process if(pid==0)//Child

Process

{
```

```
printf("\nAfter declaring FORK.");
printf("\nThe value of PID is zero.");
printf("\nThe value of PID is : ");
printf("\n%d",getpid()); printf("\nThe
value of PPID is : ");
printf("\n%d",getppid());
}
else if(pid>0)//Parent Process
{
printf("\nAfter declaring FORK."); printf("\nThe
value of PID is greater than zero."); printf("\nThe
value of PID is : "); printf("\n%d",getpid());
printf("\nThe value of PPID is : ");
printf("\n%d",getppid());
}
else//Error
{
printf("\nAfter declaring FORK.");
printf("\nThe value of PID is less than zero.");
printf("\nThe value of PID is : ");
printf("\n%d",getpid()); printf("\nThe value of
PPID is : "); printf("%d",getppid());
}

}
```

Output:



Lab2:

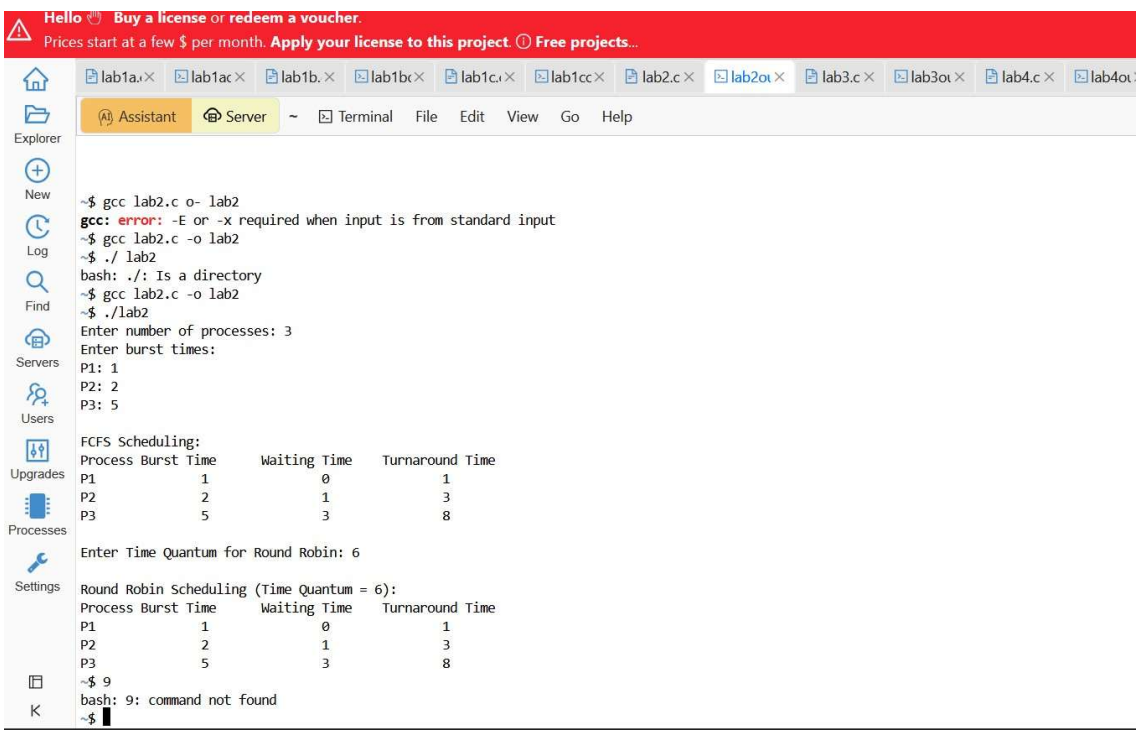
Program:

```
#include <stdio.h>  //
FCFS Scheduling
void fcfs(int n, int bt[]) {  int wt[10], tat[10];  wt[0] = 0;  for(int i = 1; i
< n; i++)  wt[i] = wt[i-1] + bt[i-1];  printf("\nFCFS Scheduling:\n");
printf("Process\tBurst Time\tWaiting Time\tTurnaround Time\n");
for(int i = 0; i < n; i++) {

tat[i] = wt[i] + bt[i];  printf("P%d\t\t%d\t\t%d\t\t%d\n",
i+1, bt[i], wt[i], tat[i]);
}
}
// Round Robin Scheduling  void
roundRobin(int n, int bt[], int tq) {  int
rem_bt[10], wt[10], tat[10];  int t = 0;
```

```
for(int i = 0; i < n; i++) {  rem_bt[i] =
bt[i];  wt[i] = 0;
}
while(1) {  int done = 1;
for(int i = 0; i < n; i++) {
if(rem_bt[i] > 0) {  done =
0;  if(rem_bt[i] > tq) {  t
+= tq;
rem_bt[i] -= tq;
            } else {
t += rem_bt[i];
wt[i] = t - bt[i];
rem_bt[i] = 0;
            }
          }
       }
       if(done == 1)
          break;
   }

   printf("\nRound Robin Scheduling (Time Quantum = %d):\n", tq);      printf("Process\tBurst
Time\tWaiting Time\tTurnaround Time\n");

   for(int i = 0; i < n; i++) {         tat[i] = bt[i] + wt[i];
printf("P%d\t\t%d\t\t%d\t\t%d\n", i+1, bt[i], wt[i], tat[i]);
   }
}
```

```
int main() {
int n, tq;      int
bt[10];

   printf("Enter number of processes: ");
scanf("%d", &n);  printf("Enter burst
times:\n");  for(int i = 0; i < n; i++) {
printf("P%d: ", i+1);  scanf("%d", &bt[i]);
}
fcfs(n, bt);  printf("\nEnter Time Quantum for Round
Robin: ");  scanf("%d", &tq);  roundRobin(n, bt, tq);
return 0;

}
```

Output:



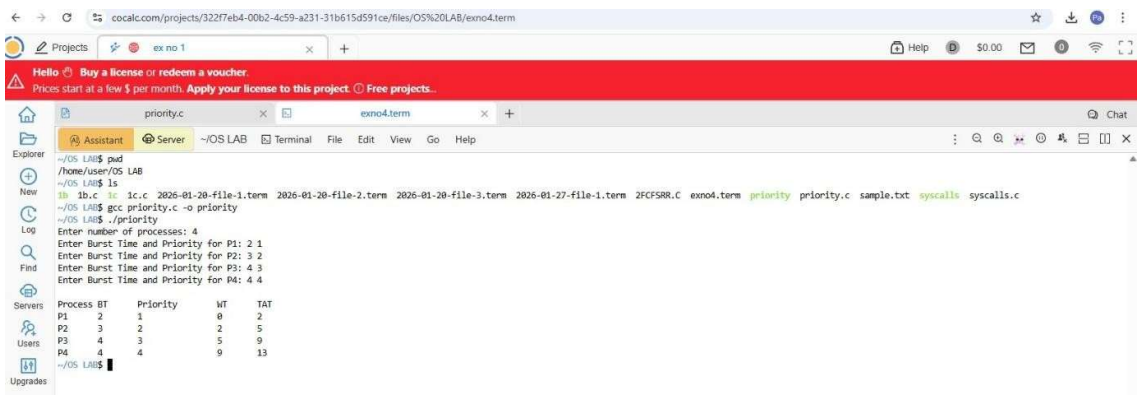
Lab3:

Program:

```
#include <stdio.h>

int main() { int
n, i, j;
int bt[10], pr[10], p[10]; int
wt[10], tat[10], temp;

printf("Enter number of processes: "); scanf("%d",
&n);

for(i = 0; i < n; i++) {
p[i] = i + 1; printf\



 Enter Burst Time and Priority for P%d: ", p[i]); scanf("%d
%d", &bt[i], &pr[i]);
}
// Sort by Priority for(i = 0; i < n; i++) {
for(j = i + 1; j < n; j++) { if(pr[i] > pr[j]) {
temp = pr[i]; pr[i] = pr[j]; pr[j] = temp;
temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
temp = p[i]; p[i] = p[j]; p[j] = temp;
}
}
```

```
}
wt[0] = 0; for(i = 1; i < n; i++) wt[i] = wt[i - 1] + bt[i - 1]; for(i = 0; i
< n; i++) tat[i] = wt[i] + bt[i];
printf("\nProcess\tBT\tPriority\tWT\tTAT\n"); for(i = 0; i < n; i++)
printf("P%d\t%d\t%d\t\t%d\t%d\n", p[i], bt[i], pr[i], wt[i], tat[i]);
return 0;

}
```

Output:



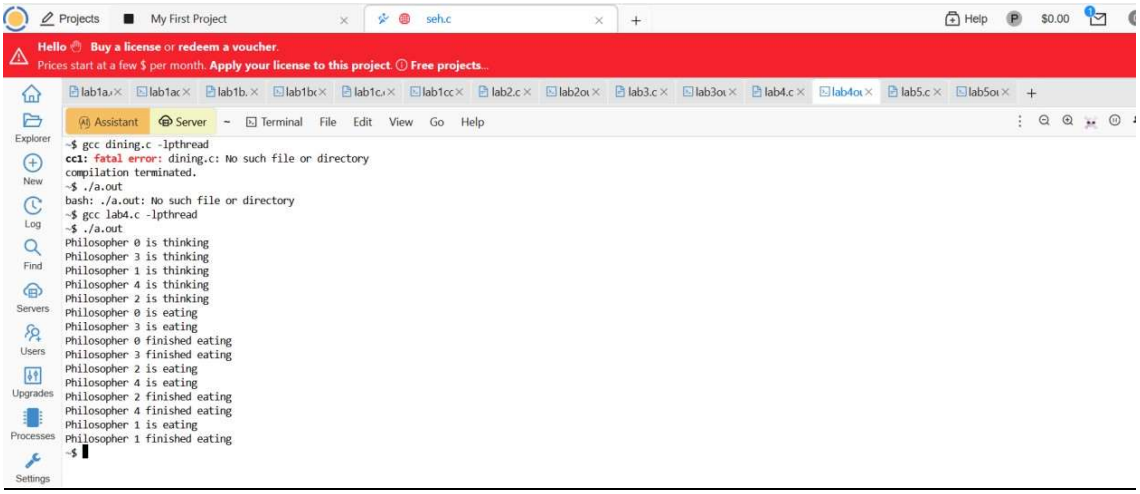
Lab4:

Program:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>  #define N 5  sem_t
forks[N];  pthread_t philosophers[N];
```

```
void* philosopher(void* num) {  int id =
*(int*)num;  printf("Philosopher %d is
thinking\n", id);  sleep(1);
sem_wait(&forks[id]);
sem_wait(&forks[(id + 1) % N]);
printf("Philosopher %d is eating\n", id);
sleep(1);   sem_post(&forks[id]);
sem_post(&forks[(id + 1) % N]);
printf("Philosopher %d finished eating\n",
id);   return NULL;
}
int main() {
int i, id[N];

   for (i = 0; i < N; i++)
sem_init(&forks[i], 0, 1);
for (i = 0; i < N; i++) {
id[i] = i;
      pthread_create(&philosophers[i], NULL, philosopher, &id[i]);
   }
   for (i = 0; i < N; i++)
pthread_join(philosophers[i], NULL);   return 0;
}
```

Output:



Lab5:

Program:

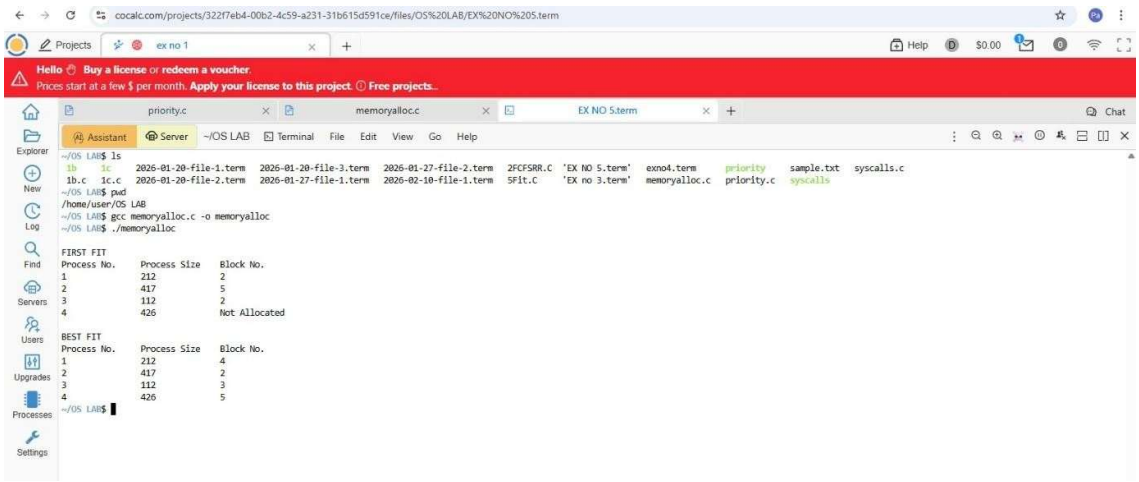
```
#include<stdio.h>  void firstFit(int blockSize[], int m, int
processSize[], int n)
{
int alloca on[n];
for(int i=0;i<n;i++)
alloca on[i]=-1;

  for(int i=0;i<n;i++)
  {
    for(int j=0;j<m;j++)
    {
      if(blockSize[j] >= processSize[i])
      {
        alloca on[i] = j;
blockSize[j] -= processSize[i];
break;
```

```
        }
    }
  }

  prin ("\nFIRST FIT\nProcess No.\tProcess Size\tBlock No.\n");
for(int i=0;i<n;i++)
  {
    if(alloca on[i]!=-1)        prin
("%d\t\t%d\t\t%d\n",i+1,processSize[i],alloca on[i]+1);    else
        prin ("%d\t\t%d\t\tNot Allocated\n",i+1,processSize[i]);
  }
}

void bestFit(int blockSize[], int m, int processSize[], int n)
{
  int alloca on[n];
for(int i=0;i<n;i++)
alloca on[i]=-1;

  for(int i=0;i<n;i++)
  {
    int bestIdx=-1;
for(int j=0;j<m;j++)
    {
        if(blockSize[j]>=processSize[i])
        {
            if(bestIdx==-1 || blockSize[j]<blockSize[bestIdx])
bestIdx=j;
```

```
        }
    }

    if(bestIdx!=-1)
    {
        alloca on[i]=bestIdx;  blockSize[bestIdx]-
=processSize[i];
    }
  }

  prin ("\nBEST FIT\nProcess No.\tProcess Size\tBlock No.\n");
for(int i=0;i<n;i++)
  {
    if(alloca on[i]!=-1)  prin
("%d\t\t%d\t\t%d\n",i+1,processSize[i],alloca on[i]+1);  else
        prin ("%d\t\t%d\t\tNot Allocated\n",i+1,processSize[i]);
  }
}

int main()
{
  int blockSize[] = {100, 500, 200, 300, 600};
int processSize[] = {212, 417, 112, 426};

  int m = 5;
int n = 4;  int
block1[5],
block2[5];
```

```
for(int
i=0;i<m;i++)
  {
      block1[i]=blockSize[i];
block2[i]=blockSize[i];
  }

  firstFit(block1,m,processSize,n);
bestFit(block2,m,processSize,n);

  return 0;
}
```

Output:



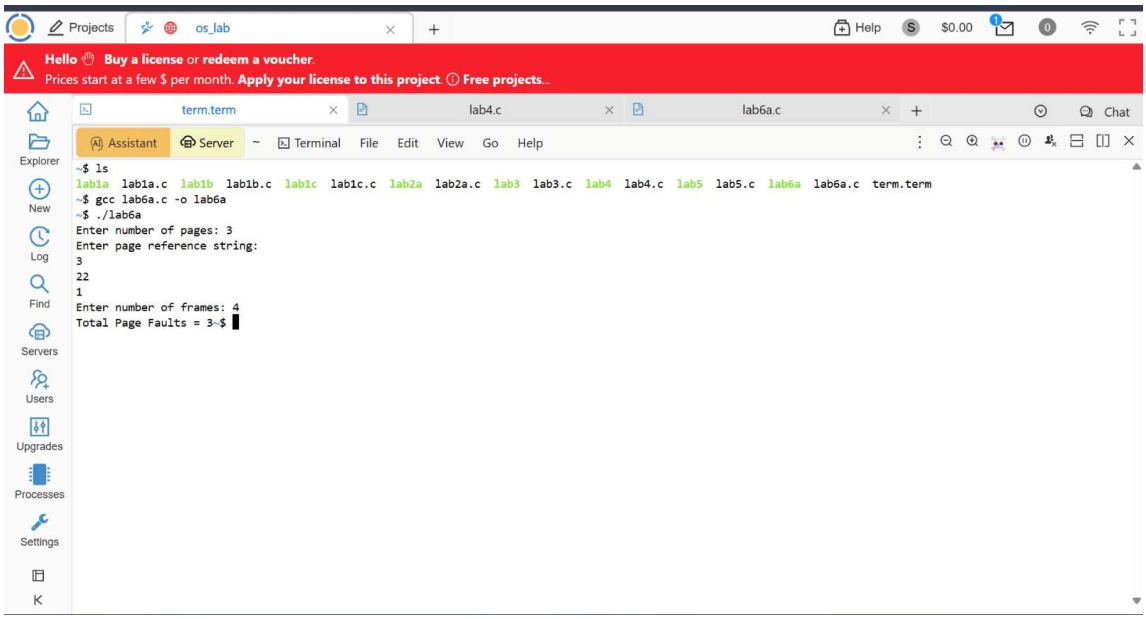
LAB 6A

PROGRAM:-

```
#include <stdio.h>
int main()
{
   int pages[50], frames[10], time[10];
   int n, f, i, j, k, flag, faults = 0, pos, min, counter = 0;

   printf("Enter number of pages: ");
   scanf("%d",&n);

   printf("Enter page reference string:\n");
   for(i=0;i<n;i++)
      scanf("%d",&pages[i]);

   printf("Enter number of frames: ");
   scanf("%d",&f);

   for(i=0;i<f;i++)
      frames[i] = -1;

   for(i=0;i<n;i++)
   {
      flag = 0;   // <-- FIXED here
      for(j=0;j<f;j++)
      {
         if(frames[j]==pages[i])
         {
            counter++;
            time[j]=counter;
```

```
        flag = 1;   // <-- FIXED here
        break;
    }
}
if(flag==0)
{
    if(i<f)
    {
        frames[i]=pages[i];
        counter++;
        time[i]=counter;
        faults++;
    }
    else
    {
        min=time[0];
        pos=0;
        for(k=1;k<f;k++)
        {
            if(time[k]<min)
            {
                min=time[k];
                pos=k;
            }
        }
        frames[pos]=pages[i];
        counter++;
        time[pos]=counter;
        faults++;
    }
}
```

```
    }
    printf("Total Page Faults = %d",faults);
    return 0;
}
```

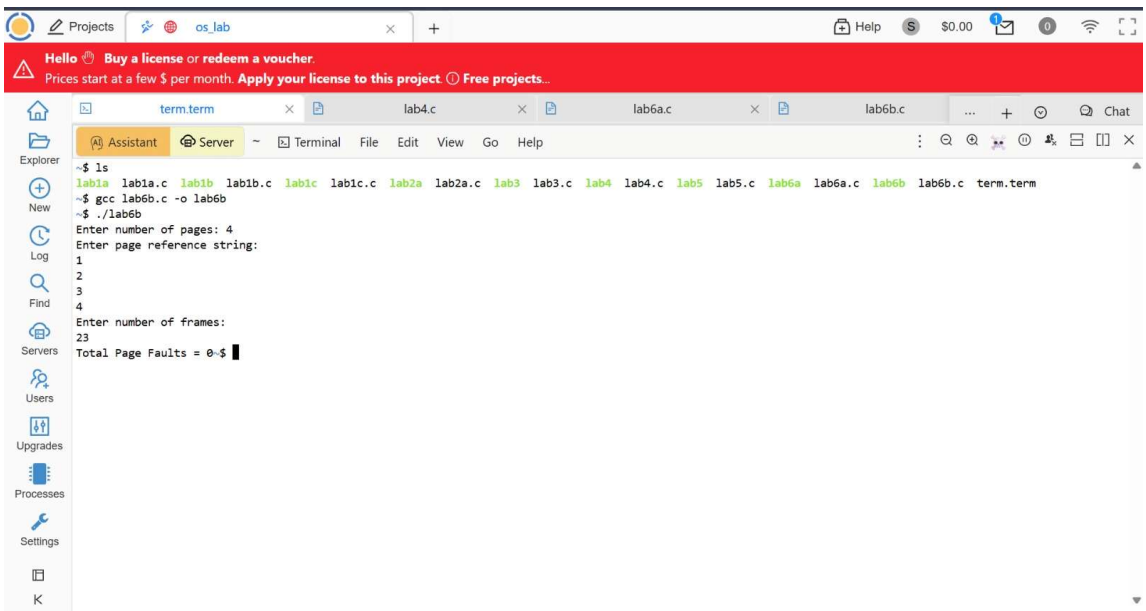


LAB 6B

PROGRAM:-

```
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, f, i, j, k, flag, faults = 0;
    int pos, farthest, index;

    // Input: number of pages
    printf("Enter number of pages: ");
    scanf("%d", &n);
```

```
// Input: page reference string
printf("Enter page reference string:\n");
for (i = 0; i < n; i++)
   scanf("%d", &pages[i]);

// Input: number of frames
printf("Enter number of frames: ");
scanf("%d", &f);

// Initialize frames to empty (-1)
for (i = 0; i < f; i++)
   frames[i] = -1;

// Process each page in the reference string
for (i = 0; i < n; i++) {
   flag = 0;

   // Check if page is already in a frame
   for (j = 0; j < f; j++) {
      if (frames[j] == pages[i]) {
         flag = 1;
         break;
      }
   }

   // If page not found (page fault)
   if (flag == 0) {
      if (i < f) {
         // Fill empty frame slots first
         frames[i] = pages[i];
```

```
    faults++;
  } else {
    // Find the page to replace (Optimal strategy)
    farthest = -1;

    for (j = 0; j < f; j++) {
      index = -1;

      // Look ahead in the reference string
      for (k = i + 1; k < n; k++) {
        if (frames[j] == pages[k]) {
          index = k;
          break;
        }
      }

      // If a page is never used again, replace it
      if (index == -1) {
        pos = j;
        break;
      }

      // Otherwise, replace the one used farthest in future
      if (index > farthest) {
        farthest = index;
        pos = j;
      }
    }

    frames[pos] = pages[i];
    faults++;
```

```
        }
      }
    }

    // Output total page faults
    printf("Total Page Faults = %d\n", faults);

    return 0;
}
```